Functional Documentation & User Guide: NY EV Site Suitability Dashboard

Target Audience: *Urban Planners, Infrastructure Developers, NYSERDA Policy Analysts, Con Ed Grid Engineers*

Interface File: *`outputs/ev_site_suitability_map.html` / `http://127.0.0.1:8000/`*

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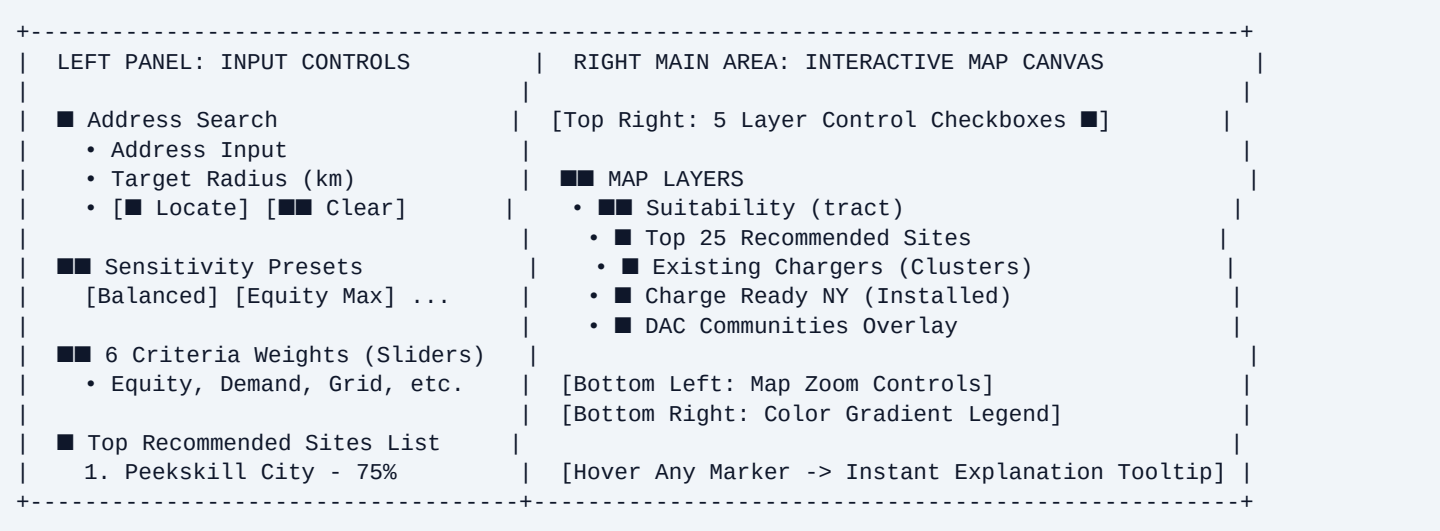
1. Interface Overview

The **NY EV Site Suitability Dashboard** features a modern **Split-Dashboard Layout**:

- **Left Control Drawer (Inputs Panel)**: Translucent glassmorphism sidebar containing search tools, scenario presets, MCDA sliders, constraint filters, and candidate site rank cards.
- **Right Map Canvas**: Full-screen interactive Leaflet map rendering base layers, suitability choropleths, existing EV chargers, Charge Ready NY projects, and DAC community boundary overlays.



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2. Step-by-Step Functional Guide

2.1 ■ Address Search & Coverage Check

Use this panel to evaluate whether any specific address in New York has adequate nearby EV charging infrastructure.

Enter Address: Type any address in the search box (e.g. *"Times Square, New York"*, *"Peekskill City Hall"*). A debounced dropdown suggests matching addresses as you type.

Set Target Coverage Radius: Enter your desired buffer distance in kilometers into the **Target Radius (km)** box (default is `2.0` km).

Click ■ `Search & Locate`:

- The map drops a pin on the searched location and flies to it at street level (Zoom level 14).

- The system measures the distance to the **nearest top-ranked candidate site**.

- **Result Status Output**:

- ■ **Bold Green (`Result: ✓ Within X km coverage radius...`)**: Appears when a top candidate site exists within your specified radius.

- ■ **Bold Red (**`Result: ✗ Outside X km coverage radius...`**)**: Appears when the location is underserved and outside the coverage radius.

Click ■■ `Clear`: Clears search text, removes distance lines and pins, and resets map zoom back to New York.

2.2 ■■ Using Sensitivity Presets

Quickly switch policy priorities using the 1-click preset buttons:

- ■■ **Balanced**: Standard 6-criteria balance (25% Equity, 20% Demand, 15% Traffic, 15% Grid, 10% Pop, 15% Gap).
- ■ **Equity Max**: Maximizes investment in NYSERDA Disadvantaged Communities (45% Equity weight).
- ■ **Demand Max**: Focuses on high EV registration density and adoption rates (35% Demand weight).
- ■ **Grid Focus**: Prioritizes Con Ed substation capacity and MW headroom (35% Grid weight).

2.3 ■■ 6-Criteria Weight Sliders

Fine-tune custom weights by dragging any of the 6 sliders:

- **Equity & DAC Status**: Prioritizes designated DAC census tracts.
- **EV Adoption Demand**: Prioritizes counties with higher registered EVs per 1,000 residents.
- **Traffic Access Volume**: Prioritizes high daily traffic corridors and highways.
- **ConEd Grid Capacity**: Prioritizes feeders with available MW load capacity.
- **Population Density**: Prioritizes densely populated urban neighborhoods.
- **Infrastructure Gap**: Prioritizes charging deserts far from existing public chargers.

Auto-Normalization: *Sliders dynamically re-normalize to 100% so scores are strictly bounded between* ***0% and 100%****.*

2.4 ■■ Layer Controls & Map Interaction

Toggle any of the 5 layer checkboxes in the top-right map menu:

- ■■ **Suitability (tract)**: Heatmap circles showing score distributions.
- ■ **Top Recommended Sites**: Numbered candidate site markers.
- ■ **Existing Chargers**: Clustered blue markers showing operational public chargers. Hovering over cluster circles shows cluster station counts.
- ■ **Charge Ready NY (installed)**: Green markers showing 1,119 completed NYSERDA state-funded EV projects.
- ■ **DAC Communities Overlay**: Full purple census tract polygon boundaries for NYSERDA Disadvantaged Communities.

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3. Obtainable Business & Analytics Outcomes

By interacting with this dashboard, stakeholders can obtain:

Site Selection Priority Ranks: Identify exact top-ranked candidate sites with complete metric breakdowns (MW headroom, population density, EV per 1k).

Infrastructure Desert Audits: Instantly determine if an address or neighborhood lacks charging access within a custom radius.

Policy Trade-off Analysis: Observe how site rankings shift when pivoting from demand-driven to equity-driven funding.

Validation Proof: Prove that model recommendations overlap with real-world state-funded *Charge Ready NY* successes.